6.2 Deformation: Elastic & Plastic Behaviour

Question Paper

Course	CIE A Level Physics
Section	6. Deformation of Solids
Topic	6.2 Deformation: Elastic & Plastic Behaviour
Difficulty	Medium

Time allowed: 20

Score: /10

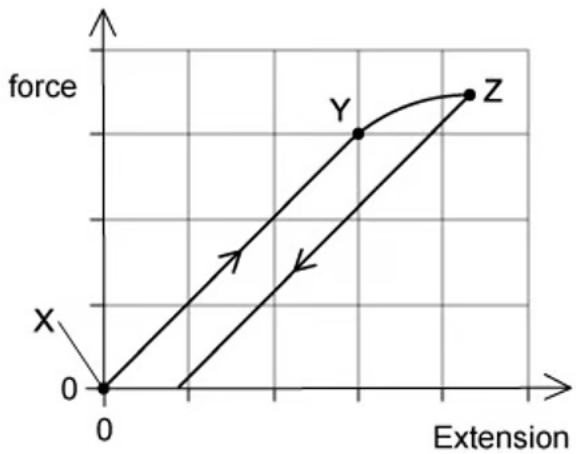
Percentage: /100

Question 1

A student was investigating the extension in a long thin metal wire. They suspended the wire from a fixed support, and hung it vertically then hung masses from the lower end.

The load was increased and decreased from zero to the maximum and back again.

The graph shows the force-extension of the wire.



Where on the graph would the elastic limit be found?

A. exactly at point Z

B. exactly at point Y

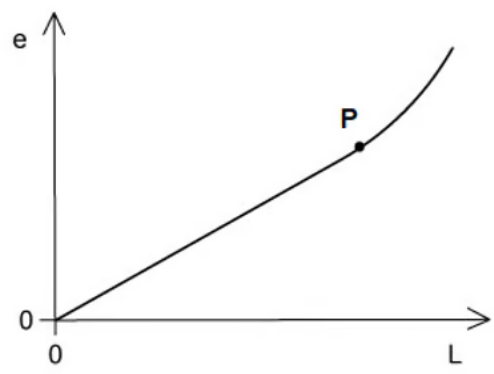
C. just beyond point Y

D. anywhere between point X and point Y

[1 mark]

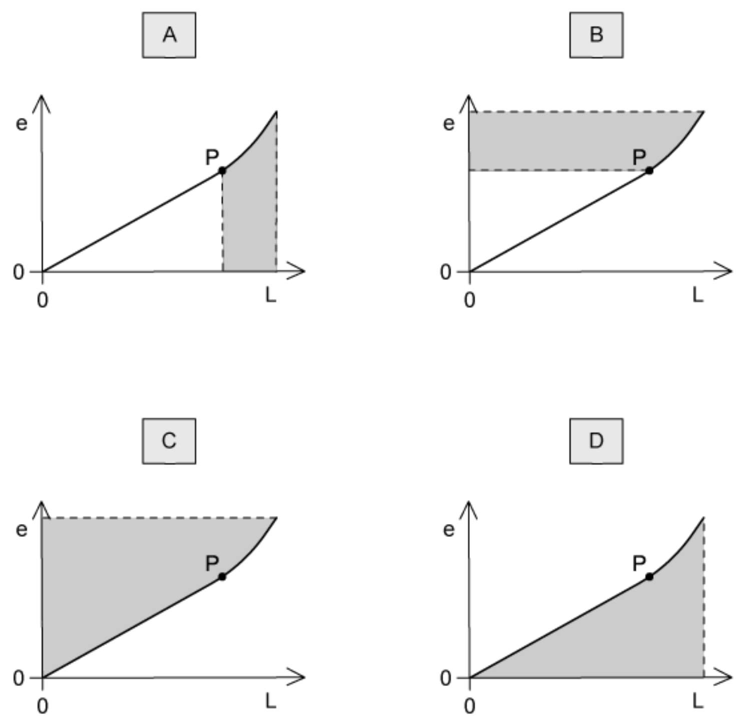
Question 2

A rod is extended with a force so that its length increases. The variation with load *L* of the extension e of the rod is shown in the graph.



Point **P** is the elastic limit.

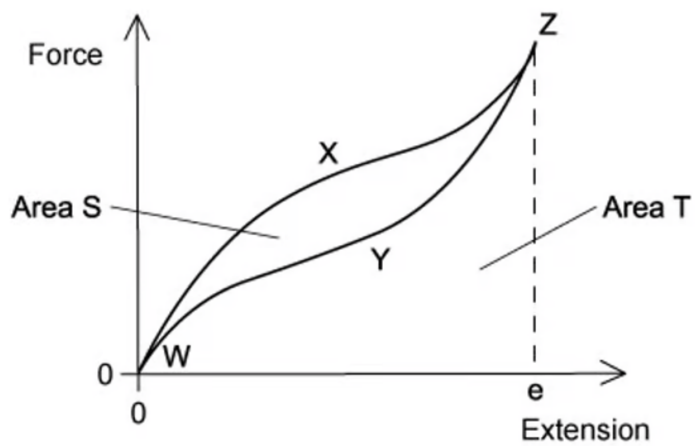
Which shaded area represents the work done during the plastic deformation of the rod ?



[1 mark]

Question 3

A student was investigating a rubber band when stretched and relaxed to its original length. The graph shows the force-extension of the rubber band.



As the force is increased, the curve follows the path WXZ to extension *x*. As the force is reduced, the curve follows the path ZYW to return to zero extension.

The area labelled S is between the curves WXZ and ZYW. The area labelled T is bounded by the curve ZYW and the horizontal axis.

Which statement about the process is correct?

A. area S is the energy which heats the band as it is stretched to e

B. area S is the elastic potential energy stored in the band when it is stretched to e

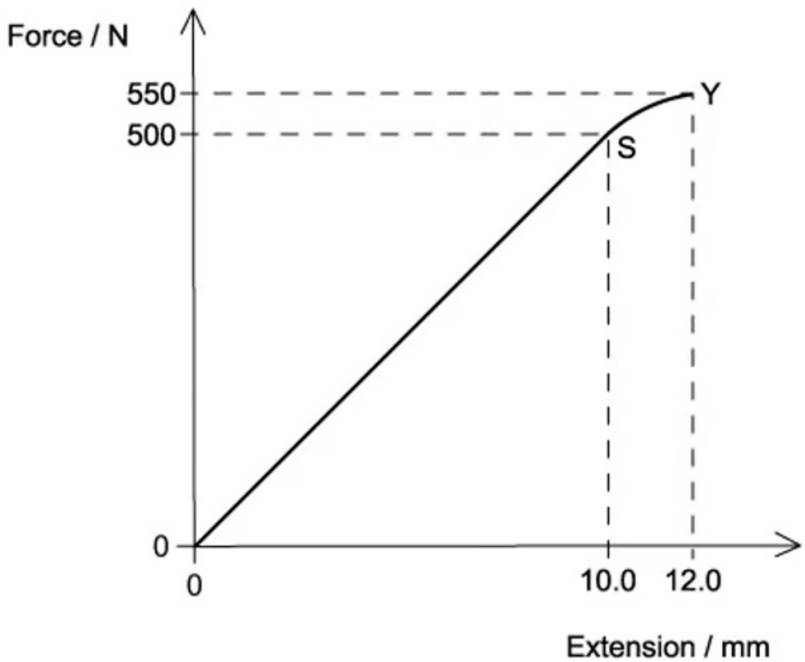
C. (area S + area T) is the minimum energy required to stretch the band to e

D. area S is the energy which heats the band as it is stretched to e

[1 mark]

Question 4

A sample of metal was stretched until it started to go into plastic deformation. The graph below shows the behaviour of the metal.



When the sample was stretched from 0 to 12.0 mm extension, what is the work done? Assume that the curve SY is a straight line to simplify the calculation

A. 6.60 J

B. 3.60 J

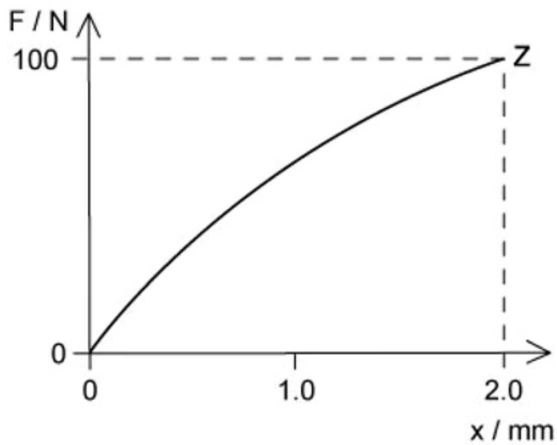
C. 3.55 J

D. 3.30 J

[1 mark]

Question 5

A wire made from a new composite material was loaded with masses and the force-extension curve drawn.



What could be the value of the strain energy stored in the wire when it is stretched elastically to point Z ?

A. 0.20 J

B. 0.11 J

C. 0.10 J

D. 0.09 J

[1 mark]

Question 6

A student was investigating a spring with a spring constant *k* of 3.0 N cm^{-1}, the spring was stretched over a range in which elastic deformation occurred.

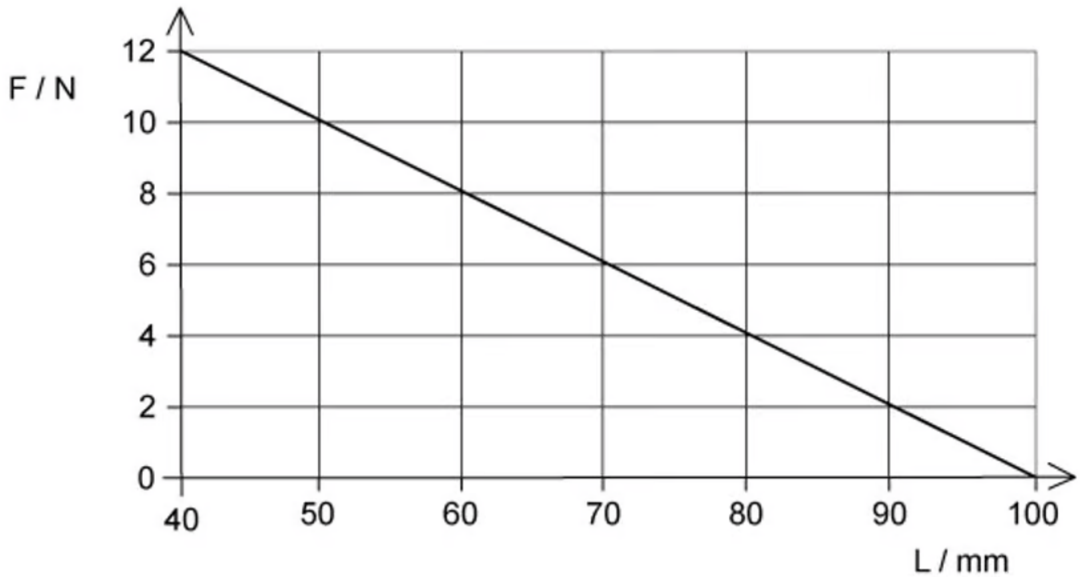
Which row, for the stated applied force, gives the correct extension and strain energy?

	Forc / N	**Extension / cm**	**Strain energy / mJ**
A	24.0	8.0	960
B	12.0	3.0	200
C	6.0	2.0	120
D	3.0	1.0	1.5

[1 mark]

Question 7

A force compresses a mattress spring by 100 mm. The graph shows the effect of the force *F* on the length *l* of the spring.



What is the energy stored in the spring when the length is 70 mm?

A. 0.63 J

B. 0.27 J

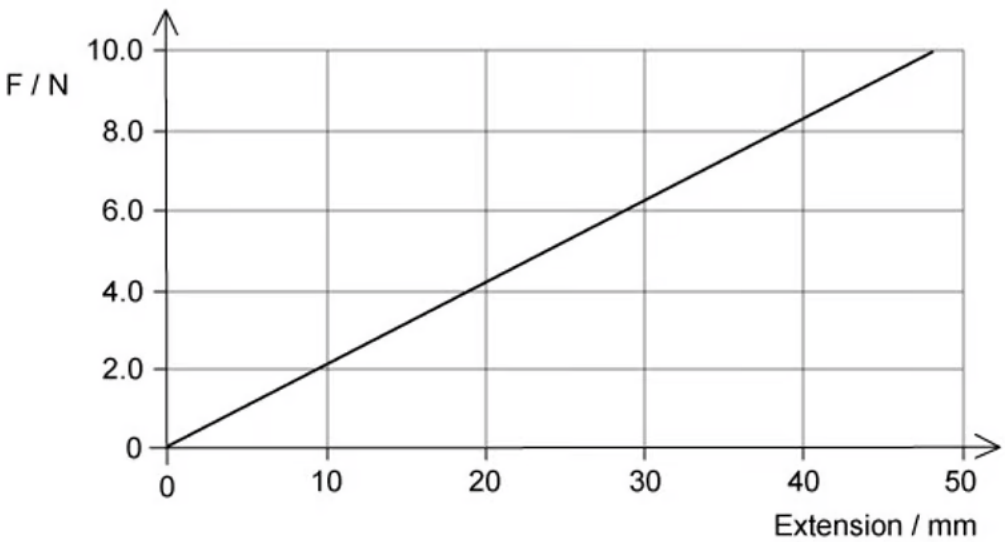
C. 0.21 J

D. 0.090 J

[1 mark]

Question 8

A spring is extended due to a force. The graph shows the effect of changing force on the extension of the spring.



What is the elastic potential energy of the spring with a force of 8.0 N ?

A. 152 mJ

B. 58 J

C. 85 mJ

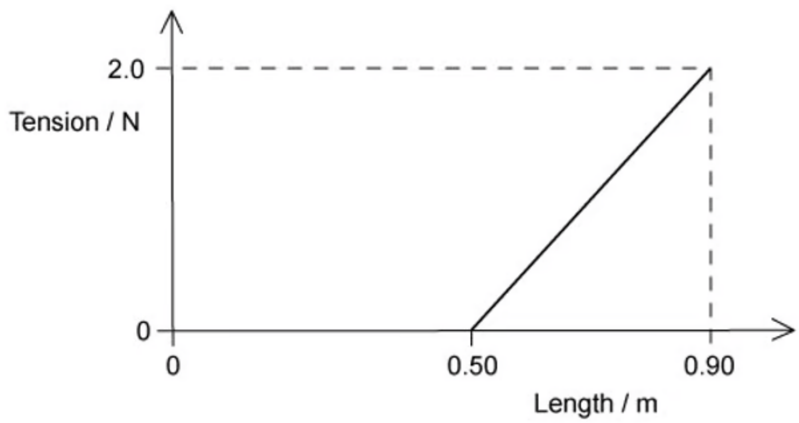
D. 5.8 mJ

[1 mark]

Question 9

A force of 2.0 N is applied to a spring of unextended length 0.50 m and has a new length of 0.90 m.

A graph shows the variation of its length with tension.



How much strain energy is stored in the spring?

A. 1.8 J

B. 0.90 J

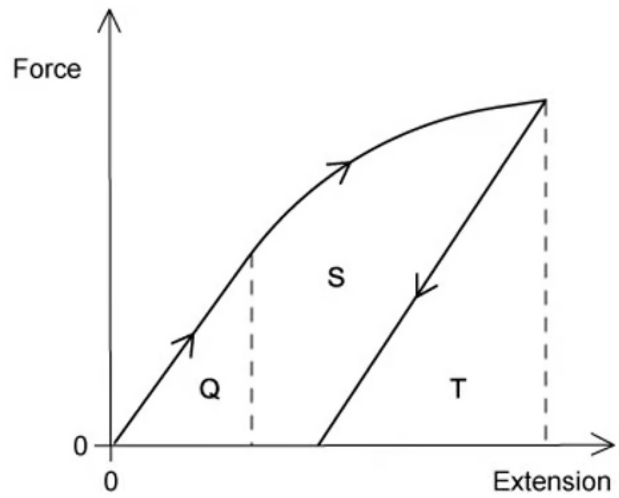
C. 0.40 J

D. 0.80 J

[1 mark]

Question 10

A student carried out an experiment on a ductile material; stretching it with a tensile force beyond its elastic limit. The force is removed.



The graph shows the force-extension of this experiment.

Which area represents the net work done on the sample?

A. T

B. $S + T$

C. $Q + S$

D. Q

[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

- The elastic limit is just beyond the limit of proportionality
  - If you increase the load beyond this point, the material will deform plastically
- The limit of proportionality is shown by point Y
  - This is the point after which Hooke's Law is no longer obeyed
- Therefore, the elastic limit is just beyond point Y

A is incorrect because point Z is beyond the elastic limit since plastic deformation occurs when the force is removed

B is incorrect because Y shows the limit of proportionality

D is incorrect because between point X and Y the force is proportional to the extension

2

The correct answer is **B** because:

- The elastic limit is shown by point P
  - This is the point at which if you increase the load any further, the material will deform plastically
- Therefore the plastic deformation is beyond point P
- The work done will be the area under the line in a load-extension graph
  - In this case, the extension *e* is on the y-axis and the load *L* is on the x-axis, so we require the area between the curve and the y-axis

A is incorrect because plastic deformation is the area under a force-extension graph not extension-force

C is incorrect because the graph shows the work done for the whole extension not just plastic deformation

D is incorrect because work done is the area under a force-extension graph not extension-force

Exam Tip: always check the axes of the graph, many exam questions will change the axes to a form you are not used to in order to catch you out, so look out!

3

The correct answer is **C** because:

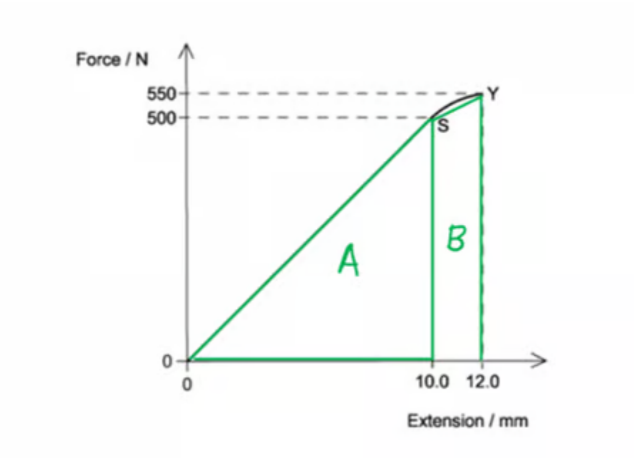
- The area under the curve OXZ is the work done in stretching the rubber band
    - Area S represents the amount of thermal energy dissipated to the surroundings
    - Area T represents the energy that is stored at maximum extension
- Therefore, the minimum energy required to stretch the band to *e* is equal to = (area S + area T)

A, B & D are incorrect because the area S represents the amount of thermal energy dissipated to the surroundings by the rubber band

4

The correct answer is **C** because:

- Work done = force × distance = area under the graph
- Divide the total shape into a triangle and a trapezium



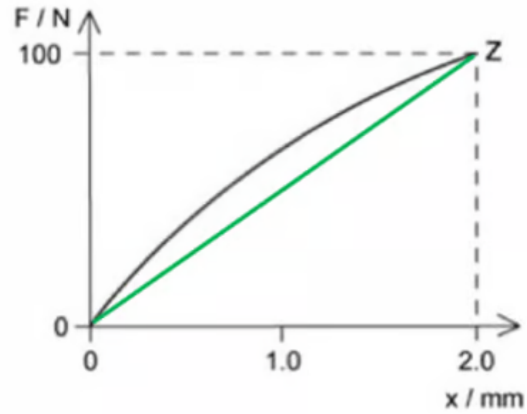
- Area of shape A = $\frac{1}{2} \times 500 \times 0.010 = 2.50J$

- Area of shape B = $\frac{1}{2} \times (500 + 550) \times 0.002 = 1.05J$
  - Therefore, total area $= 2.50 + 1.05 = 3.55J$

5

The correct answer is **B** because:

- The area under a force-extension curve gives strain energy
- Draw a straight line drawn from (0, 0) to point Z



- Calculate the area of the triangle
  - Strain energy $= \frac{1}{2}Fx = \frac{1}{2} \times 100 \times (2 \times 10^{-3}) = 0.1J$
- This is less than the area under curve shown in the graph
- The answer should be about 10% greater to account for the curve = 0.11 J

A is incorrect because this is twice the calculated value so too big

C & **D** are incorrect because the actual area under the curve should be greater than the figure calculated

6

The correct answer is **A** because:

- Using Hooke's law $F = kx$
  - Where k is the spring constant, 3.0 N cm^{-1}, or 300 N m^{-1}

- Strain energy = $\frac{1}{2}Fx$
- Consider option **A**:
  - Force = 24 N, extension = 0.08 m
  - Strain energy $= \frac{1}{2} \times 24 \times 0.08 = 0.96J = 960\ mJ$

B is incorrect because force = 12 N, extension = 0.03 m

strain energy = 0.5 × 12 × 0.03 = 0.18 J = 180 mJ

C is incorrect because force = 6 N, extension = 0.02 m

strain energy = 0.5 × 6 × 0.02 = 0.06 J = 60 mJ

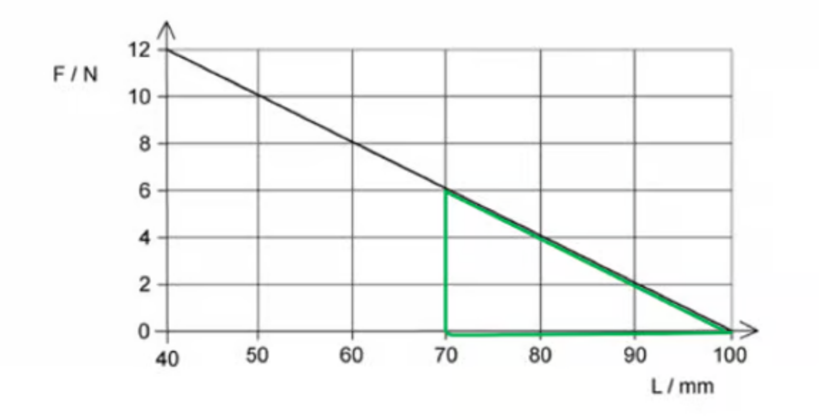
D is incorrect because force = 3.0 N, extension = 0.01 m

strain energy = 0.5 × 3 × 0.01 = 0.015 J = 15 mJ

7

The correct answer is **D** because:

- The original length of the spring is 100 mm this is shown on the graph where the force is zero
- The energy stored when the spring is compressed to 70 mm is equal to the area under the graph
  - Therefore, compression = 100 – 70 = 30 mm = 0.03 m



strain energy = 0.5 × 6 × 0.02 = 0.06 J = 60 mJ

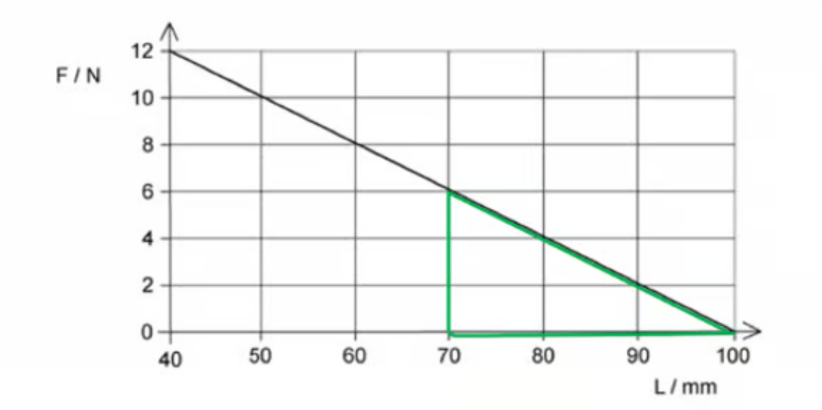
D is incorrect because force = 3.0 N, extension = 0.01 m

strain energy = 0.5 × 3 × 0.01 = 0.015 J = 15 mJ

7

The correct answer is **D** because:

- The original length of the spring is 100 mm this is shown on the graph where the force is zero
- The energy stored when the spring is compressed to 70 mm is equal to the area under the graph
  - Therefore, compression = 100 – 70 = 30 mm = 0.03 m



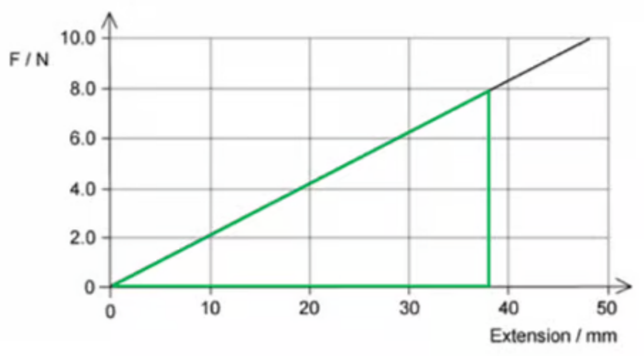
- Elastic potential energy = $\frac{1}{2}Fx$
  - From the graph:
  - When compression = 100 – 70 = 30 mm = 0.03 m
  - Force = 6 – 0 = 6.0 N
- Therefore, strain energy $= \frac{1}{2} \times 6.0 \times 0.03 = 0.090 J$

8

The correct answer is **A** because:

- The elastic potential energy is the area under the force-extension graph.

- The question asks for the elastic potential energy when a force of 8.0 N is applied



- Elastic potential energy $= \frac{1}{2}Fx$
- Elastic potential energy $= \frac{1}{2} \times 8.0 \times 0.038 = 0.152J \text{ or } 152\ mJ$

9

The correct answer is **C** because:

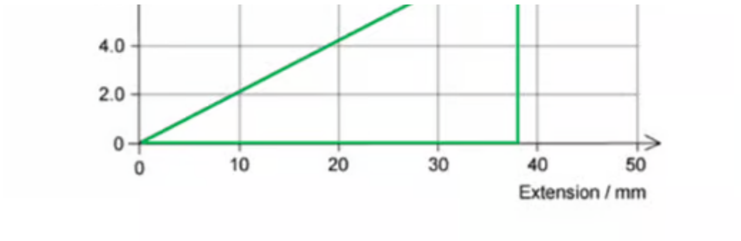
- Energy stored in a spring is equal to $= \frac{1}{2}Fx$
- The area under the tension-length graph gives the energy stored in the spring
  - Therefore, strain energy $= \frac{1}{2} \times (2.0 - 0) \times (0.9 - 0.5) = 0.40J$

Exam Tip: Strain is equal to $\frac{extension}{original\ length}$ this is not the same as the energy stored in the spring

10

The correct answer is **C** because:

- Strain energy $= \frac{1}{2}Fx$



- Elastic potential energy $= \frac{1}{2}Fx$
- Elastic potential energy $= \frac{1}{2} \times 8.0 \times 0.038 = 0.152J \text{ or } 152 \text{ } mJ$

9

The correct answer is **C** because:

- Energy stored in a spring is equal to $= \frac{1}{2}Fx$
- The area under the tension-length graph gives the energy stored in the spring
  - Therefore, strain energy $= \frac{1}{2} \times (2.0 - 0) \times (0.9 - 0.5) = 0.40J$

Exam Tip: Strain is equal to $\frac{extension}{original\ length}$ this is not the same as the energy stored in the spring

10

The correct answer is **C** because:

- Strain energy $= \frac{1}{2}Fx$
  - So, the area under the curve in a force-extension graph equals the work done
- *Q*, *S* and *T* show sections of the area under the line
- The materials behaviour is shown as the force is applied and then removed
  - Therefore, to calculate the work done the total area enclosed by the curve is needed so *Q* + *S*